



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMSTER 2 2025/2026

SECI1143 PSDA

SECTION 02

ASSIGNMENT 2

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Question 1

1. a) sample space = $\{E, S, P, D\}$

b) (i) $A \cup B$

$$= \{E, S, D\}$$

(ii) $A \cap B$

$$= \emptyset$$

c) (i) $P(A) = \{E, S\}$

(ii) $P(B^c) = \{E, P, D\}$

Question 2

2. a) $P(X=x) = 1$

$$P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) = 1$$

$$0.1 + 0.25 + 0.3 + 0.2 + d = 1$$

$$d = 0.150$$

b) (i) $P(X \leq 2)$

$$= P(X=0) + P(X=1) + P(X=2)$$

$$= 0.10 + 0.25 + 0.30$$

$$= 0.650$$

(ii) $P(1 < X \leq 3)$

$$= P(X=2) + P(X=3) = 0.30 + 0.20 = 0.500$$

c) (i) $\sum_{x=0}^4 x P(x) = 0.1(0) + 0.25(1) + 0.3(2) + 0.2(3) + 0.15(4) = 2.050$

$$(ii) \text{Var}(X) = \sum x^2 P(x) - (\sum x P(x))^2$$

$$= 0.1(0)^2 + 0.25(1)^2 + 0.3(2)^2 + 0.2(3)^2 + 0.15(4)^2 - 2.05^2$$

$$= 1.448$$

Question 3

Tarikh

No.

a) X is the number of batteries that fail the quality test in a random sample of 4 batteries ($n=4$).

b) Binomial Distribution. $X \sim B(4, 0.2)$.

$$\begin{aligned} \text{c) i. } P(X=2) &= {}^4C_2 (0.2)^2 (1-0.2)^2 \\ &= {}^4C_2 (0.2)^2 (0.8)^2 \\ &= 0.1536 \\ &= 0.154 \end{aligned}$$

$$\begin{aligned} \text{ii. } P(X \leq 1) &= P(X=0) + P(X=1) \\ &= {}^4C_0 (0.2)^0 (0.8)^4 + {}^4C_1 (0.2)^1 (0.8)^3 \\ &= 0.4096 + 0.4096 \\ &= 0.819 \end{aligned}$$

$$\begin{aligned} \text{d) mean, } \mu &= np \\ &= 4(0.2) \\ &= 0.800 \end{aligned}$$

$$\begin{aligned} \text{variance, } \sigma^2 &= npq \\ &= 4(0.2)(0.8) \\ &= 0.640. \end{aligned}$$

Question 4

(a) Y represent the number of products inspected until the first defective product is found.

(b) geometric distribution.

$$\begin{aligned} \text{(c) (i)} \quad P(X=5) &= (1-0.1)^{5-1} (0.1) \\ &= (0.9)^4 (0.1) \\ &= 0.06561 \\ &= 0.066 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(X=1) &= (1-0.1)^{1-1} (0.1) = 0.1 \\ P(X=2) &= (1-0.1)^{2-1} (0.1) = 0.09 \\ P(X=3) &= (1-0.1)^{3-1} (0.1) = 0.081 \\ P(X=4) &= (1-0.1)^{4-1} (0.1) = 0.0729 \end{aligned}$$

$$\begin{aligned} P(X \leq 4) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= 0.1 + 0.09 + 0.081 + 0.0729 \\ &= 0.3439 \\ &= 0.344 \end{aligned}$$

Question 5

$$a) i. P(X < 46) = P\left(Z < \frac{46 - 40}{6}\right)$$

$$= P(Z < 1)$$

$$= 0.841$$

$$ii. P(34 < X < 52) = P\left(\frac{34 - 40}{6} < Z < \frac{52 - 40}{6}\right)$$

$$= P(-1 < Z < 2)$$

$$= P(Z < 2) - P(Z < -1)$$

$$= 0.97725 - 0.15866$$

$$= 0.819$$

$$b) P\left(Z > \frac{X - 40}{6}\right) = 0.05$$

$$P\left(Z < \frac{X - 40}{6}\right) = 0.95$$

$$\frac{X - 40}{6} = 1.645$$

$$X = 49.87 \text{ minutes}$$

Question 6

(a) negative binomial distribution

$$\begin{aligned} (b) \quad \sigma &= \sqrt{\frac{r(1-p)}{p^2}} \\ &= \sqrt{\frac{3(1-0.6)}{0.6^2}} \\ &= \sqrt{\frac{10}{3}} \\ &= 1.8257 \\ &= 1.826 \end{aligned}$$

$$\begin{aligned} (c) \quad b^*(5; 3, 0.6) &= \binom{5-1}{3-1} (0.6)^3 (1-0.6)^{5-3} \\ &= {}^4C_2 (0.6)^3 (0.4)^2 \\ &= 0.207 \end{aligned}$$